

EAI-421

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

FOOD QUALITY GRADING OF PROCESSED FOODS

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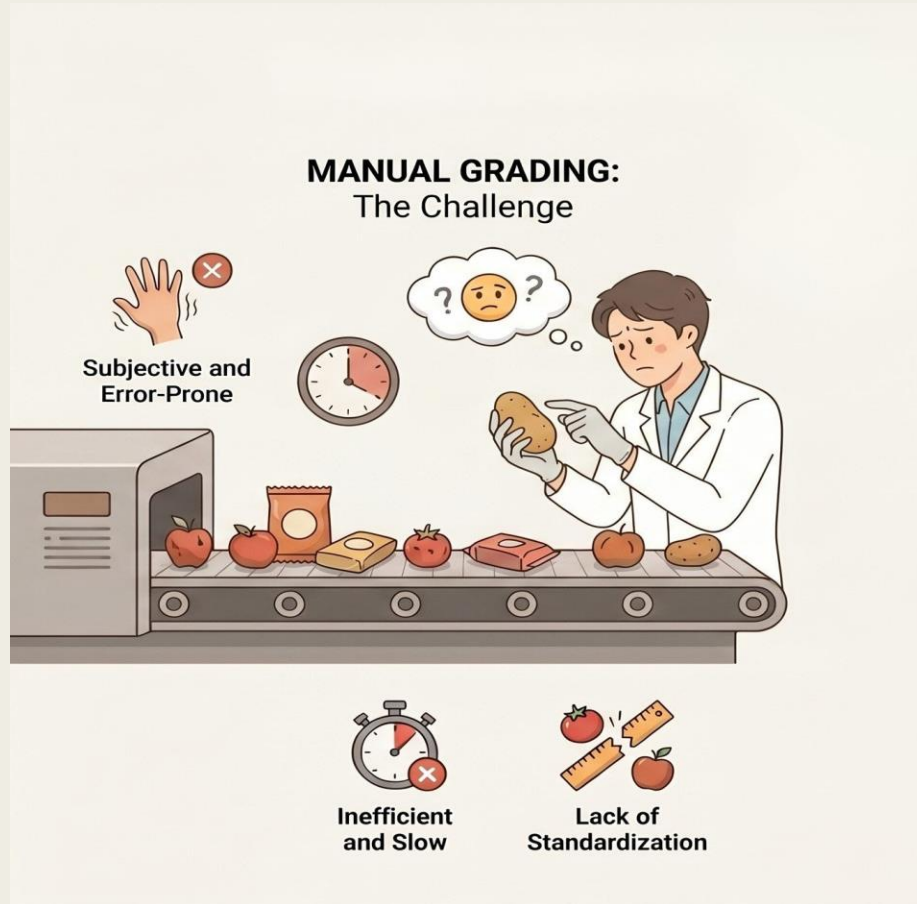
INTRODUCTION

Defining Food Quality Grading

Food quality grading is the systematic evaluation of processed foods based strictly on safety and quality standards.



PROBLEM STATEMENT



costly and time consuming



labor-intensive



inconsistent and prone to human error



Lack of standardization



Problem of real-time quality assessment

Project Aim: Develop a system for accurate, fast, and reliable food quality grading

MOTIVATION

1. Growing demand for safe and high-quality processed foods.
2. Limitations of manual grading methods (slow, subjective, error-prone).
3. Need for consistent and standardized quality assessment
4. Increasing role of AI/ML in food industry automation
5. Opportunity to apply image processing and data analysis in real-world problems
6. Supports.
7. Compliance with standards set by Food Safety and Standards Authority of India
8. Enhances efficiency, accuracy, and cost-effectiveness.
9. Contributes to consumer safety and trust

OBJECTIVES OF THE PROJECT

1. Develop an automated grading system using AI/ML.
2. Improve accuracy and consistency in food quality assessment.
3. Enable real-time and cost-effective analysis.
4. Reduce manual effort and errors.

DATASET OVERVIEW

Dataset Architecture & Input Feature Space

SOURCES

Kaggle Food Safety Portal | FSSAI Datasets | UCI Repository

VOLUME

50 Samples

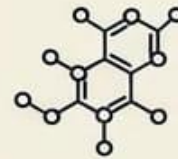
ARCHITECTURE

9 Total Features
(8 Inputs, 1 Target)



Chemical Parameters

- Moisture Content
- pH Level



Nutritional Profile

- Fat %
- Protein %



Sensory Metrics

- Colour Score
- Texture Score

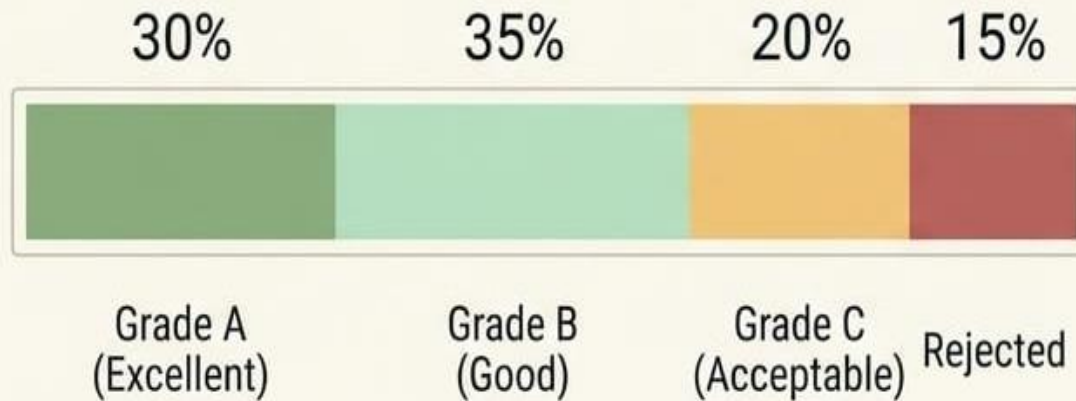


Lifecycle Data

- Shelf Life (days)
- Contamination Index

Target Output Distribution & Predictive Drivers

Target Variable
(Supervised Learning Multi-class Classification)



Key Predictive Insights

Contamination Index

Shelf Life (days)

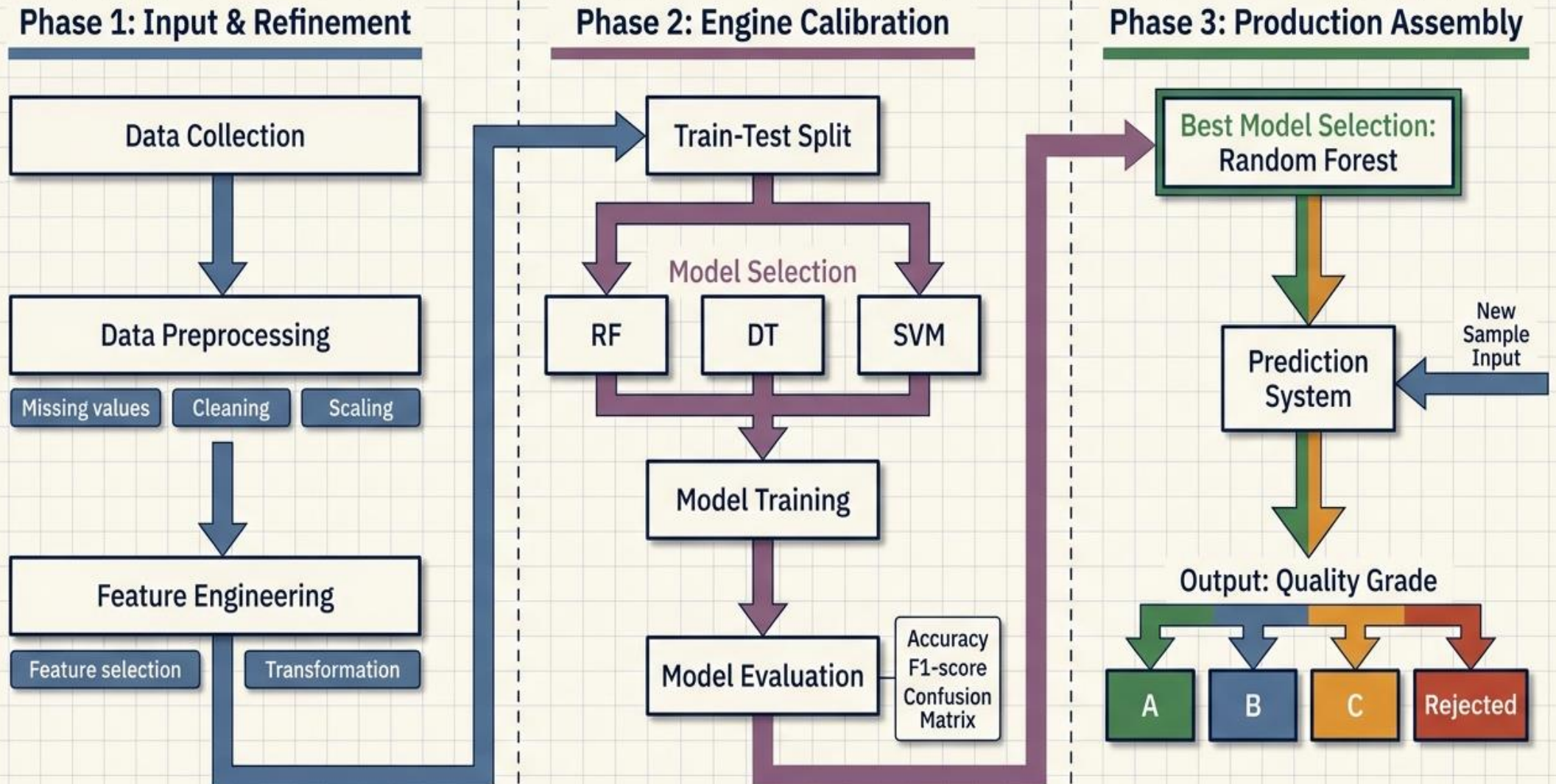
Colour Score

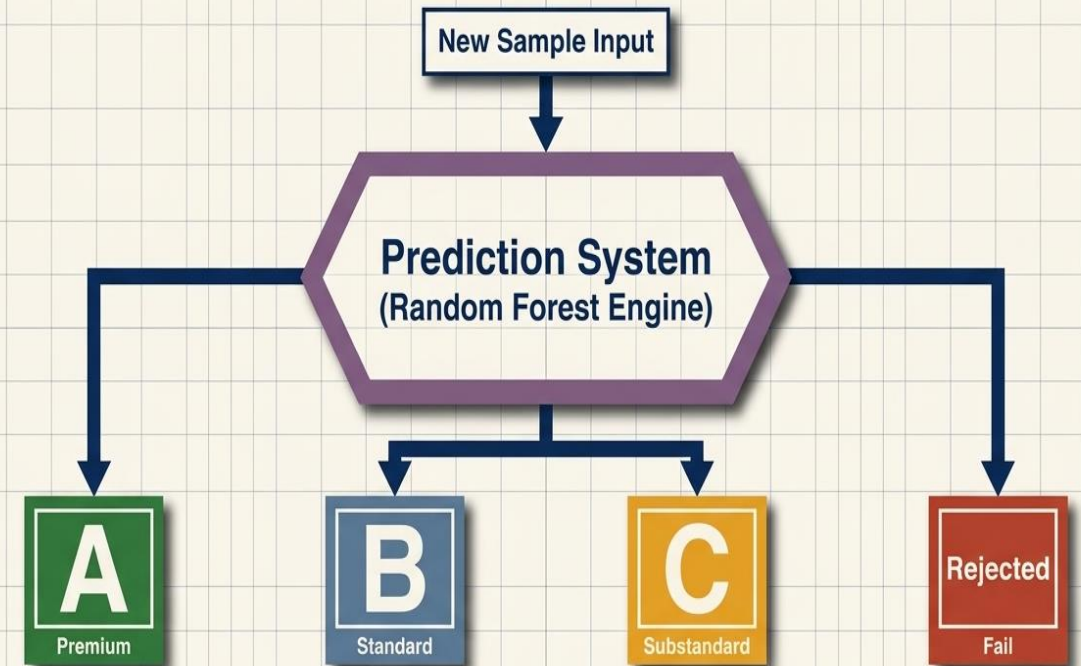
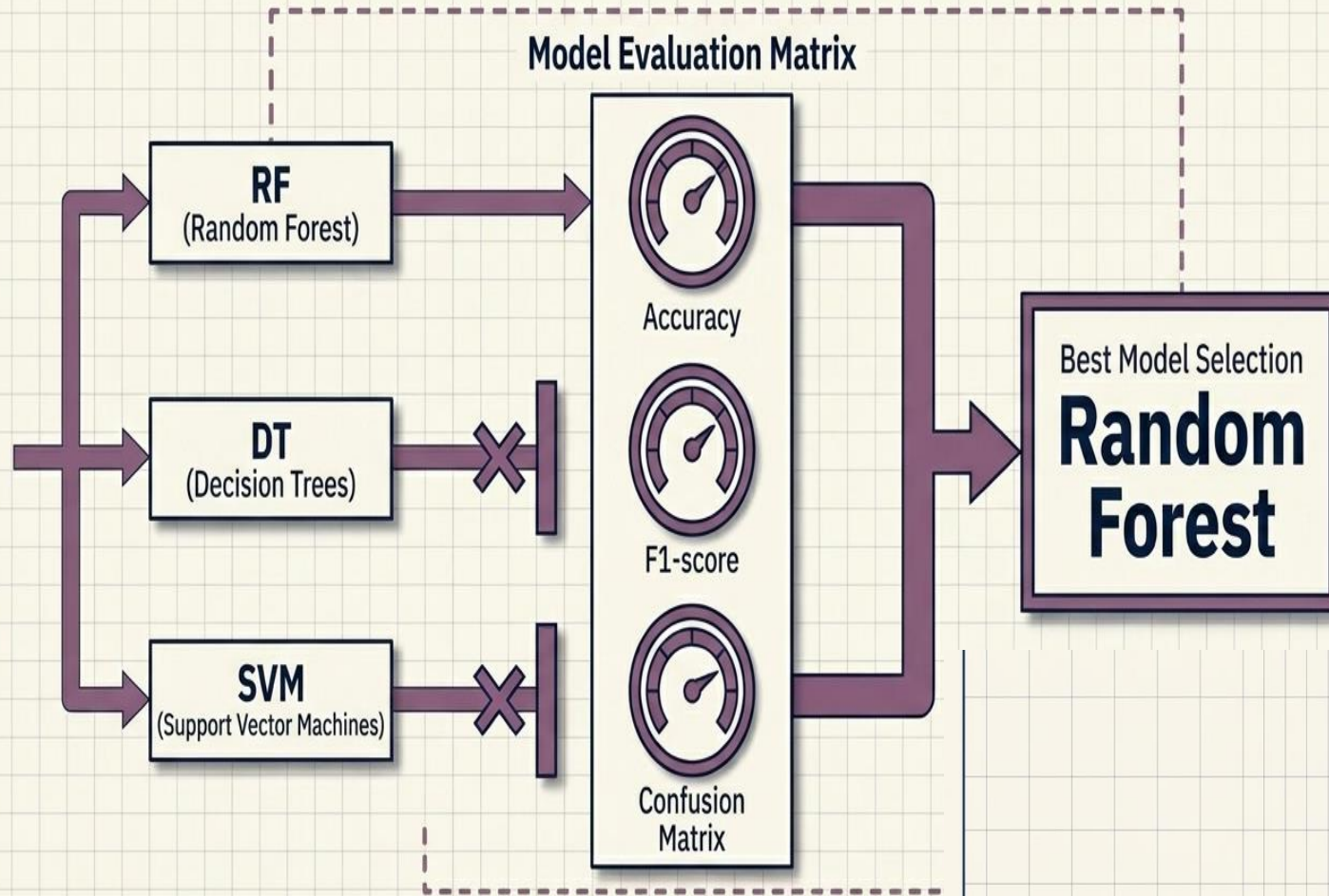
These three variables operate as the major factors influencing overarching food quality.

Dataset

	Moisture (%) (Moisture %)	Feature-2 (pH Level)	Feature-3 (Color Score)	Feature-4 (Shelf Life (Days))
	2.5%	6.2	85	120
	35%	5.5	70	5

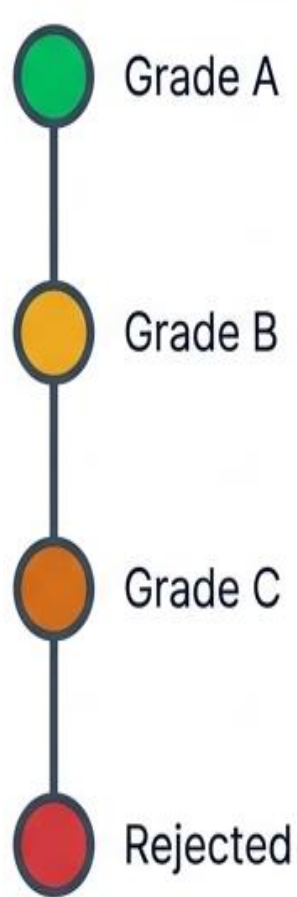
METHODOLOGY/APPROACH





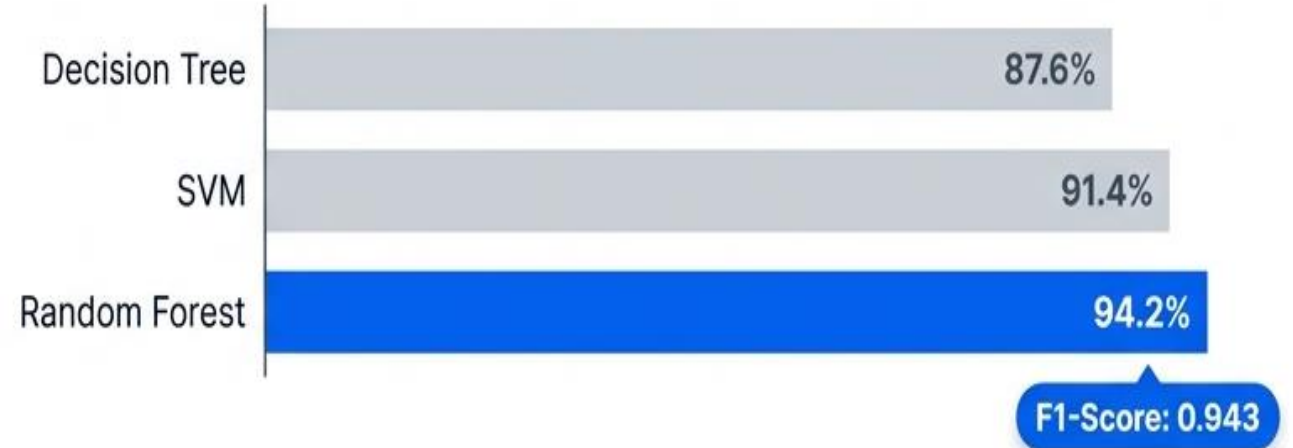
RESULTS AND DISCUSSION

Quality Spectrum



Consistent performance across the spectrum: **F1-Scores** remain **above 90%** for all classes.

Model Performance Comparison



Insight Flags



The **Contamination Index** acts as the most important feature driving quality grading.



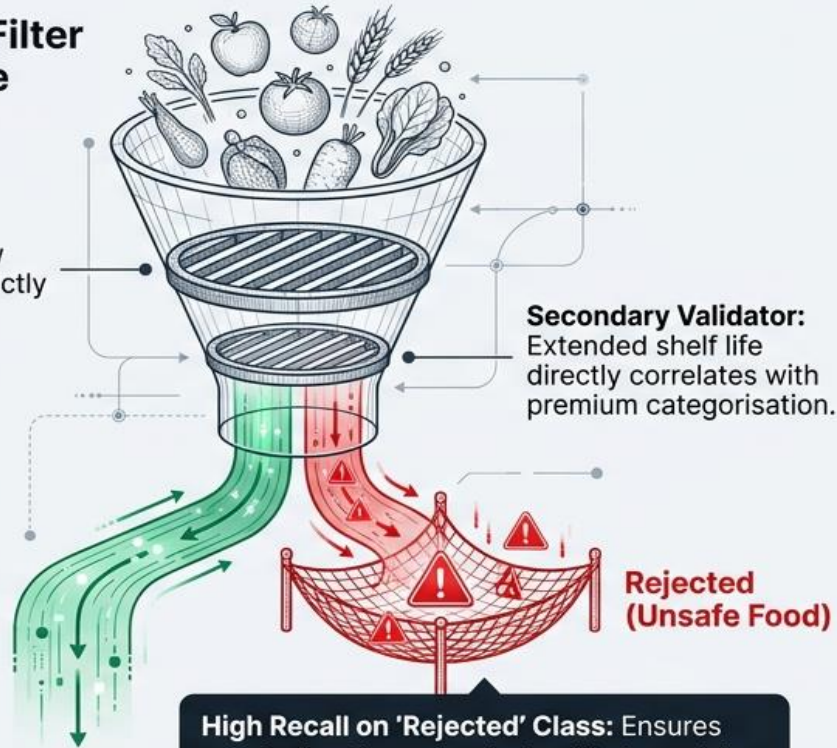
Low-risk error profile: Minor misclassifications occur almost exclusively between adjacent grades (Grade B and Grade C).

DISCUSSION AND ANALYSIS

High-Precision Food Grading: The Random Forest Engine

The Safety Filter Architecture

Primary Quality Determinant: Low contamination strictly dictates superior grading.



Random Forest vs. Alternative Models

- ✓ **Handles Non-linear Data:** Captures complex, real-world variables easily.
- ✓ **Handles Imbalanced Data:** Maintains high accuracy even with skewed datasets.
- ✓ **Computationally Efficient:** Significantly lower computational cost compared to SVM models.
- ✓ **Highly Interpretable:** Provides clear feature importance with minimal overfitting.

Scientific Synthesis

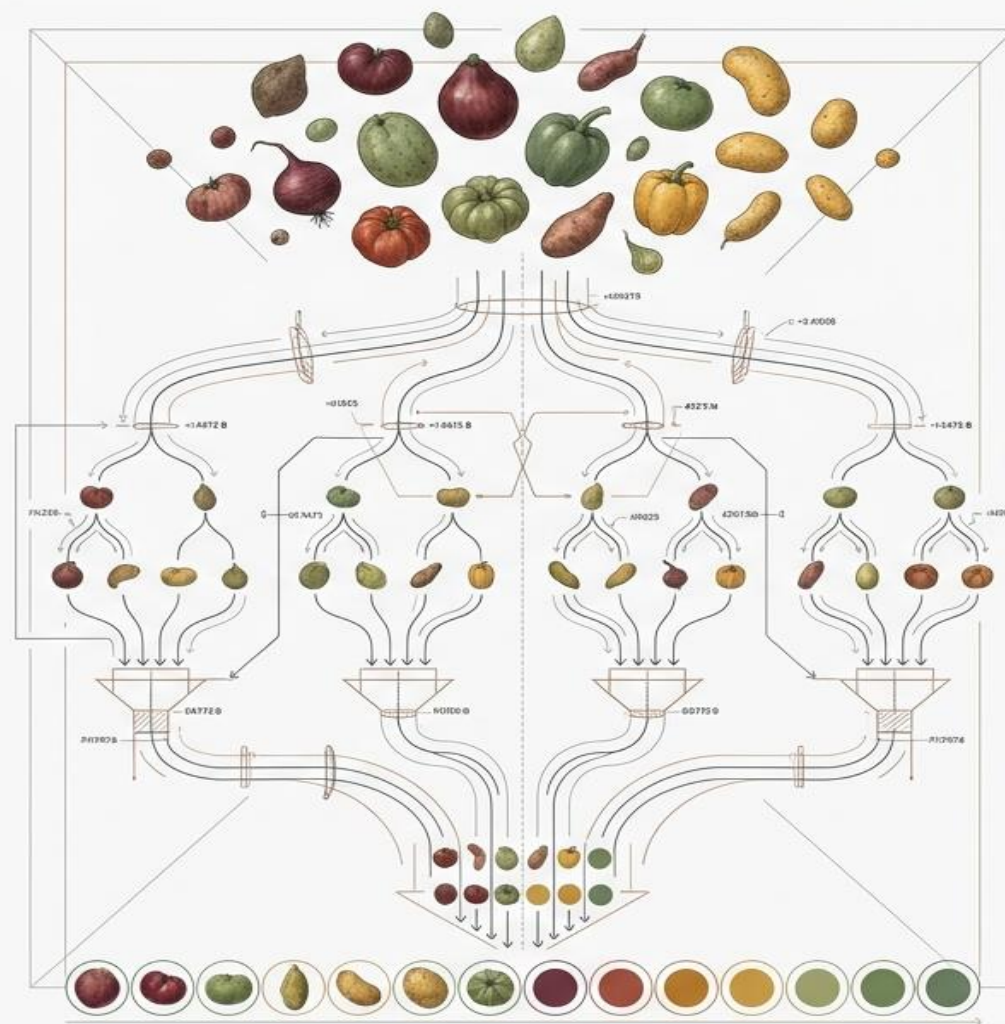
Why it works: The model's interpretable feature importance maps perfectly to established food science. By isolating contamination and shelf life as non-linear variables, the Random Forest engine creates a highly robust grading baseline.

CONCLUSION

Random Forest modelling achieves definitive grading accuracy

94.2%

AI and machine learning successfully automate food quality grading. Out of all tested algorithms, the Random Forest model proved to be the best-performing solution.



Operational velocity and
precision guarantee
consumer protection



Faster
grading

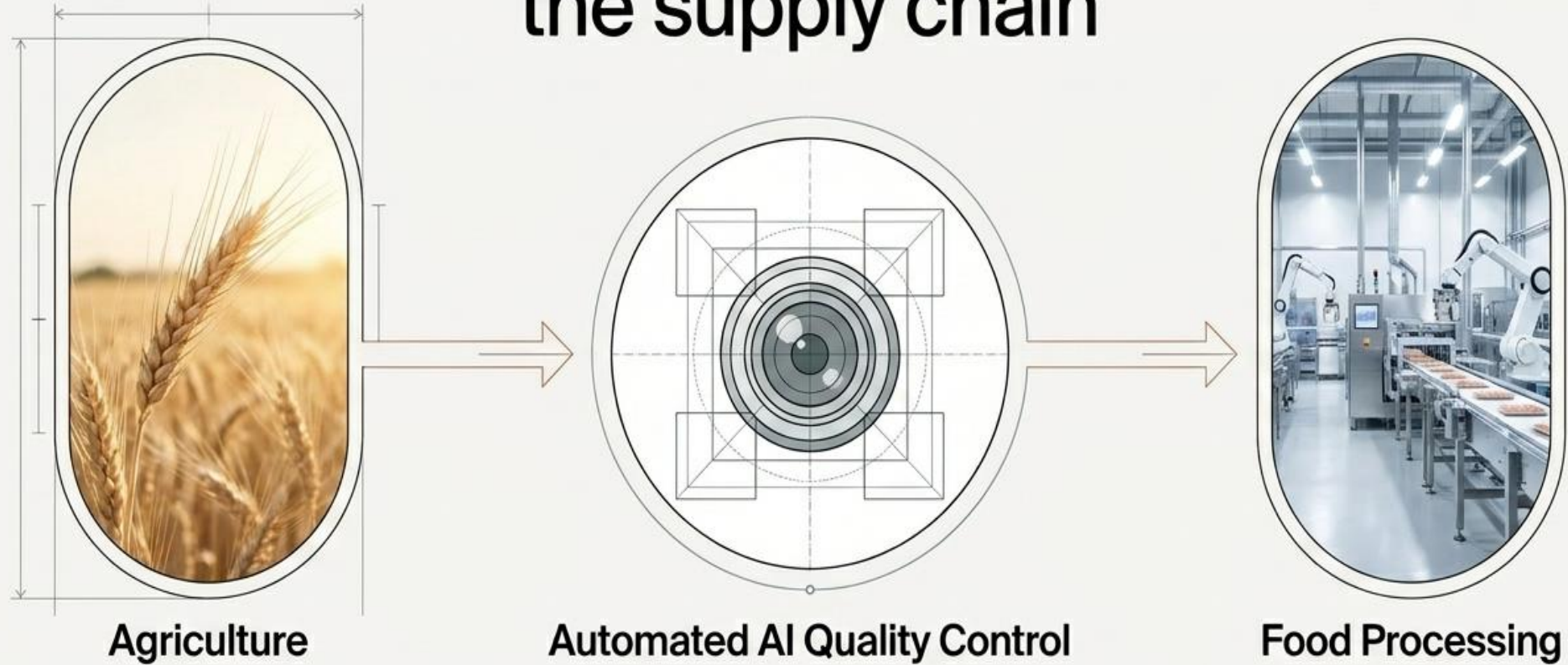


Reduced
human error



Improved
food safety

Scaling automated control across the supply chain



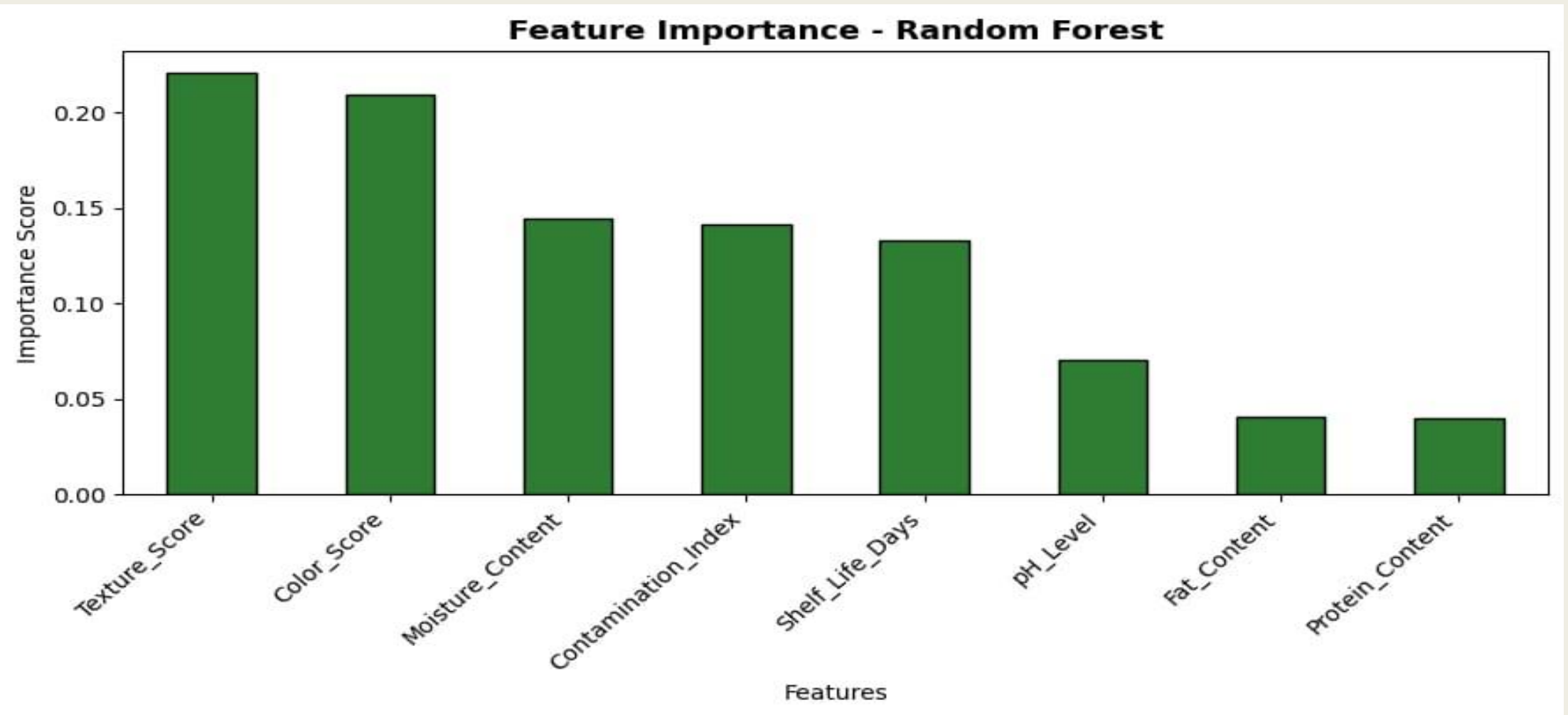
This project proves the overarching potential of AI in transforming food quality control systems across both the agriculture and food processing industries.

REFERENCES

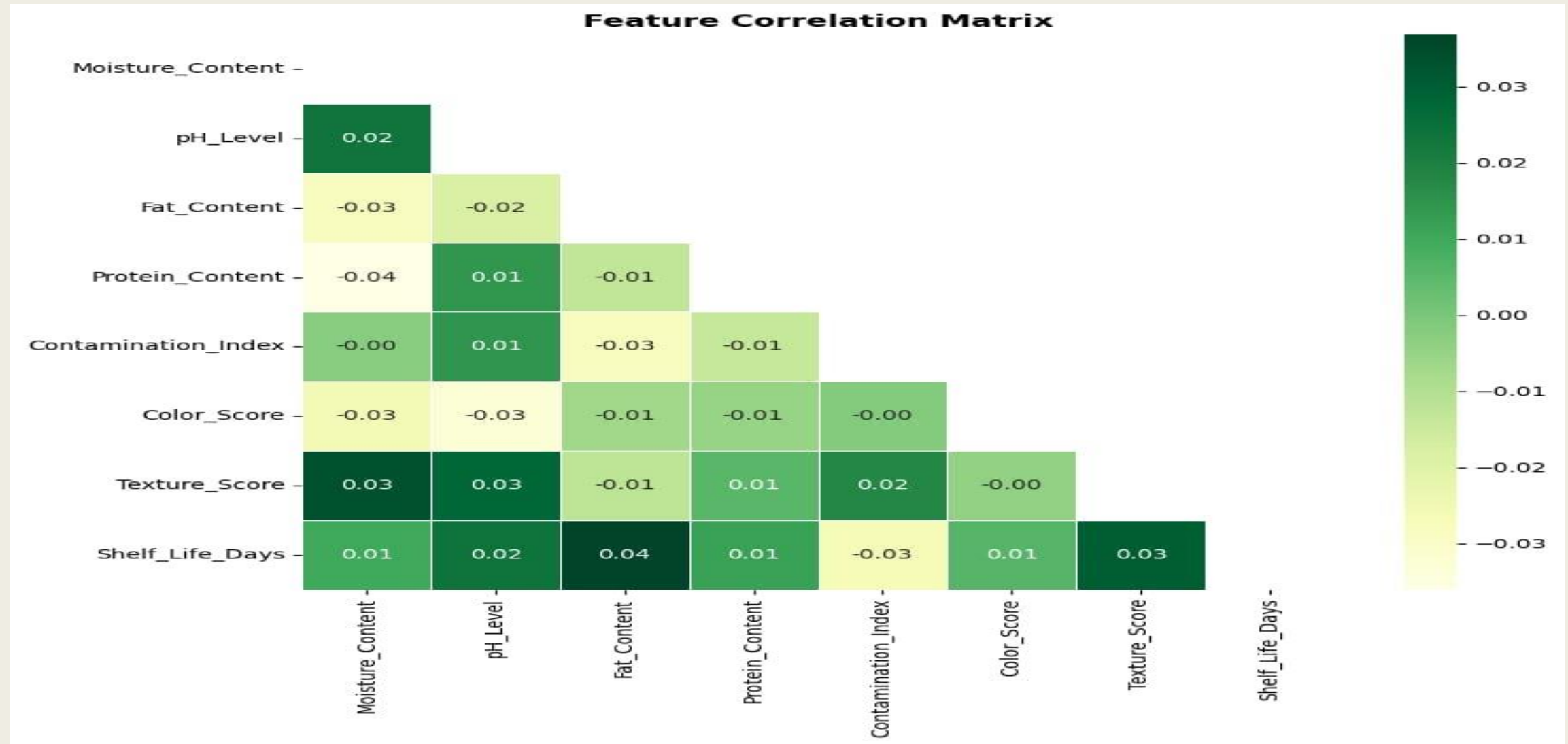
- >Kim, T. Y., et al. (2021)-Machine Learning-Based Food Quality Grading System.
- >FSSAI (2023)-Food Safety Datasets – Government of India.
- >Kaggle (2024)- Food Quality Dataset.
- >Breiman, L. (2001)- Random Forests. Machine Learning Journal.
- >Government of India (2023)- Agricultural Statistics Report

GRAPHS





How food looks and feels matters more than its nutrition for grading.



No strong relationships exist between features. Each factor (moisture, fat, protein, contamination, etc.) independently affects food quality. This is useful for models like Random Forest because they can learn from multiple independent inputs effectively.

FOOD QUALITY GRADER



Food Grader

Enter physicochemical parameters of your food sample to receive an instant AI-based quality grade classification.



SAMPLE PARAMETERS

MOISTURE & CHEMICAL

MOISTURE CONTENT

%

60.1

PH VALUE

PH

10.7

PROTEIN CON

Higher protein → better grade

%

10.7

FAT CONTENT

%

19.6

SUGAR, SODIUM & MICROBIAL

SUGAR CONTENT

%

3.8

THANK
YOU.....